

cryptographically link the past events to the current events, thus preserving their order under the latest time stamp.

Cryptocurrency does not require an intermediary like a bank to transact money. Instead, it makes use of blockchain technology. It has been in news, unfortunately, for enabling illegal purchases, money laundering, and tax evasion, which has created a wrong perception in the society. The legitimacy of dealing with this digital currency was difficult as there were no hard and fast rules to unveil the credentials of the involved criminals. However, after following the rules and guidelines set by various governments and the international Financial Action Task Force (FATF), cryptocurrency exchanges seem to be now able to keep the phenomenon under control.

Tourism industry

Planning travel often confuses and tires us due to the complexity and the time involved in buying air tickets, managing transfers, booking rentals, hotels, and many more involved activities. Use of blockchain can bring safety and transparency to a plethora of organisations that are involved in a travel plan. A traveller is required to show passport or a valid ID at multiple stages of his journey such as booking, boarding, and check-in at the hotel. Blockchain could generate a single travel token for each traveller and this information would be universally accessible to multiple airports, thus making the ID verification process simple, secure, and efficient.

Making finance settlements between travel agents, hotels, and other travel-related firms is an extremely complicated process, especially when it comes to overseas transactions that involve currency exchange. Blockchain can streamline this process by reducing the intermediaries that can intervene or delay the settlement timing and increase the overall transparency and the trust to a whole new level.



Figure 1: AI increases global GDP (Image credit: pwc.com)

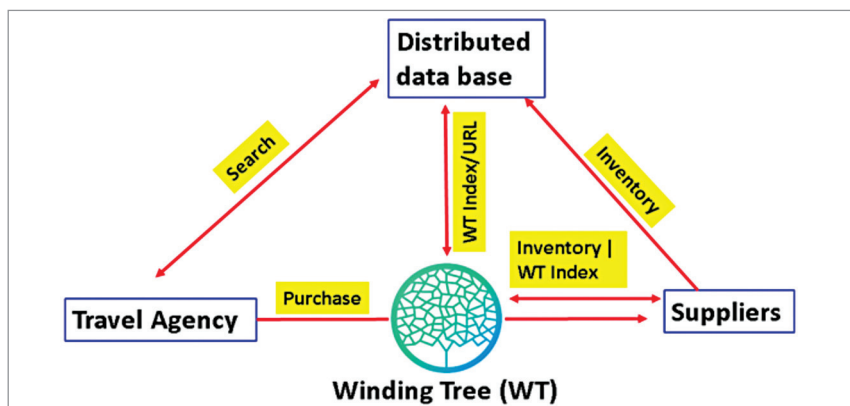


Figure 2: Winding Tree decentralised ecosystem

Baggage tracking is another challenge that requires keeping track of the travellers' luggage as it changes hands from airlines to the ground handling services at the airports. The luggage data can be stored and processed by blockchain that allows passengers to track their luggage over the cloud and thus reduce baggage mishandling. Blockchain will allow all parties to exchange information even if they are not related to the same airline or same subdivision, thus improving the accuracy of baggage tracking across the industry.

Blockchain in use

Winding Tree is a decentralised open source business-to-business (B2B) travel marketplace. It is a platform where buyers and sellers of travel inventory may find trusted partners they could not find before, connect directly, and start doing business immediately. Winding Tree is not owned by anyone; it is controlled by the community of its participants. It uses different licences for each category. For example, for smart contracts it uses AGPL (Affero General Public License), for libraries and code

examples it is licensed by Apache 2.0, and for all the texts found in the platform it is licensed by the Creative Commons Attribution 4.0 International. In short, any establishment can install this technology and the infrastructure.

Winding Tree uses Ethereum technology, a public blockchain that is best suitable for smart contracts, which provides greater cost-effectiveness for travel industry partners. Eventually, customers and other stakeholders involved receive a better deal. Disintermediation, that is, cutting off the middlemen, reduces the total cost of servicing as Winding Tree acts as a bridge between the business and their customers. The high-profile airline groups that have signed deals with Winding Tree aim to eliminate intermediaries and the accompanying costs to create a decentralised marketplace. These include Air Canada, Air France, KLM, Air New Zealand, Austrian Airlines, Brussels Airlines, Etihad Airways, Eurowings, Germany based Hahn Air, Lufthansa, Swiss Air, and Swissport.